

Chapter 9

Friday, July 19, 2024

9:37 AM

Electron Density = Mass density * $\sum (a_i \left(\frac{Z_i}{A_i} \right))$ where Z is atomic number, A is atomic weight, and a is the weighting factor.

- Solid water is an Epoxy resin-based mixture, mass density is 1.00 and number of electrons per gram: 3.34×10^{23} (perfectly matching water).

PDD: $P = D_d / D_o \times 100$ where D_o is the reference dose, and D_d is the dose to evaluate at.

- $D_{max} = D_d / P \times 100$
- PDD is effected by beam energy, inverse square law, scatter, and beam attenuation.



Brook notes

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$$\rho_e = \rho_m \cdot N_A \cdot \sum a_i \left(\frac{Z_i}{A_i} \right)$$

↑ electron density ↑ mass density ↑ atomic number (charge) ↓ atomic weight

$PDD = \frac{D_d}{D_o} \times 100\%$

↑ dose of interest ↑ reference dose (typically D_{max})

- attenuation, inverse square, scatter





Depth in Water

Kerma $K = \frac{dE_{tr}}{dm}$ Energy transferred from photons to directly ionizing electrons

↳ decreases uniformly to represent a

DOSE → Builds up as high speed electron production decreases in 2nd electrons are projected downward

Field Size

Geometric Field size

Most common

Geometrically, it's the source proj. given by light field at SSD/SD



Dosimetric/Physical Field size Distance intercepted by isodose line. (50%)

Equivalent fields
a-square
→ $a = \frac{A}{P}$ (for comparing rectangles to square)

square length

$$r = \frac{1}{\sqrt{2}} \frac{A}{P}$$

radius



- High energy beams are less affected by field size.
- PDD increases w/ higher SSDs. (inverse square reduces intensity while making the PDD less def.)

$$P(d, r_d) = 100 \cdot \left(\frac{f_1 + d_m}{f_1 + d} \right)^2 e^{-\mu(d-d_m)} \cdot \frac{K_s}{K}$$

← $f_1 + d$ ← $f_1 + d_m$ ← μ ← K_s ← K

$$f_1 = \text{SSD}_1 \quad \text{inverse square attenuation scatter}$$

$$f_2 = \text{SSD}_2 \quad //$$

Mayneord F-factor

$$\frac{P(d, r, f_1)}{P(d, r, f_2)} = \left(\frac{f_2 + d_m}{f_1 + d_m} \right)^2 \left(\frac{f_1 + d}{f_2 + d} \right)^2$$

Put 2 on top for d_{max}

Mayneord Factor:

Put unknown & d_{max} on top.

$$\frac{P(d, r, f_1)}{P(d, r, f_2)} = \left(\frac{f_2 + d_m}{f_1 + d_m} \right)^2 \left(\frac{f_1 + d}{f_2 + d} \right)^2$$

Ex: $\frac{P(10, 15, 100)}{P(10, 15, 80)} = \left(\frac{100 + 0.5}{80 + 0.5} \right)^2 \left(\frac{80 + 10}{100 + 10} \right)^2 = 1.043$

known known d_{max} d_{ref}

$$P(10, 15, 100) = 1.043 \cdot 58.4 = 60.9$$

known to overestimate increased SSD conditions, slightly higher number due to reduced fall off.

TAR

$$TAR(d, r_d) = \frac{D_d}{D_{fs}} \Rightarrow \text{dose at a given point in phantom} \Rightarrow \text{dose at some point in free space}$$

- energy
- depth
- field size

only a function of depth, energy, & field size
no reliance on SSD or scatter

$$TAR(d, 0) = e^{-\mu(d - d_m)}$$

without scatter, the tail looks exponential

$$BSF = \frac{D_{max}}{D_{fs}} \text{ so } TAR(d_{max}, r_d) = BSF$$

BSF is TAR under special conditions that you are at distance w/ reference field size rd.

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TAR & PDD

$$P(d, r, f) = \text{TAR}(d, r_f) \cdot \frac{1}{\text{BSF}} \left(\frac{f + d_{\text{cm}}}{f + d} \right)^2$$

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$$\frac{P(d, r, f_2)}{P(d, r, f_1)} = \frac{\text{TAR}(d, r, f_2)}{\text{TAR}(d, r, f_1)} \times$$

$$\left[\left(\frac{f_1 + d}{f_2 + d} \right)^2 \left(\frac{f_2 + d_{\text{cm}}}{f_1 + d_{\text{cm}}} \right)^2 \right]$$

$$\text{SAR}(d, r_d) = \text{TAR}(d, r_d) - \text{TAR}(d, 0)$$

↑
Primary component.

